

4.2 - Related Rates

Goals: Relate changing quantities by differentiating equations with respect to time.

Before Class Videos: 4.2.1

Example 1: An inverted conical tank has radius 6 ft and height 8 ft. Water is pumped in at $4 \text{ ft}^3/\text{min}$. How fast is the water level rising when the water is 4 ft deep?

Example 2: A rocket rises vertically at 600 m/s . An observer stands 4000 m from the launch point. How fast is the angle of elevation changing when the rocket is 3000 m high?

Method:

Example 3: Two objects are 400 km apart horizontally. Their vertical separation is 295 km and is increasing at 15 km/h. How fast is the distance between them changing?

Example 4: A point moves along the curve $y = x^2$, with $x > 0$. Find the rate of change of its distance from $(3, 0)$ in terms of x and dx/dt .

Example 5: Suppose a bar of length d leans against a wall and slides along the ground. If $x'(t)$, the rate of change of its horizontal position is constant, is $y'(t)$ constant?

Example 6: Suppose a $2m$ tall skateboarder is a distance x from a $5m$ tall wall which has a light at the top. If $x'(t) = 2m/2$, how fast is the length of the shadow cast by the light increasing?

Example 7: A trough is 20 m long, with triangular ends 5 m high and 5 m wide. Water flows in at $100 \text{ m}^3/\text{h}$. How fast is the water level rising when the water is 2 m deep?